Project INSPIRE Design Document

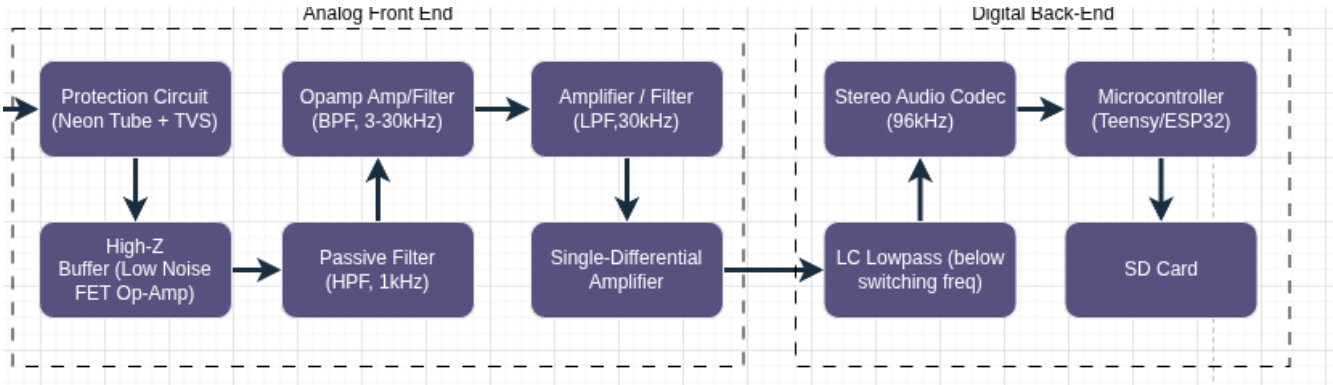
Project Part VII: ADC Conditioning

Revision 1.0: Presented 10/9/2026

1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:



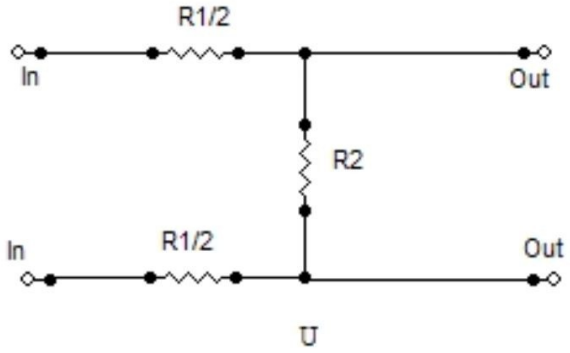
The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

2) Sub-circuit Specifications

The analog conditioning circuit protects the ADC from overvoltage, which is especially important given pro-level audio equipment has 12V standard. This will destroy the ADC if it was tied to the input, which means that an attenuator circuit is required if the recorder is to be dual-purpose.

A voltage divider in differential signaling requires a “U-bridge” circuit, where the shunt resistor goes between the differential lines. This means, we can very easily create an adjustable voltage attenuator by splitting our series resistor in half and using a switch to set the path between the differential legs.



This, combined with a ADC driver IC, can provide us with both the desired attenuation and the constant-current driving a modern ADC requires.

- **Attenuation Requirements:** The network must be calculated to drop professional line-level signals (+4 dBu / ~1.23 Vrms, with peaks up to 12V) down to safe consumer line-level or ADC-compatible windows (-20 dBV reduction). When set to "unity," the shunt path is left open, presenting zero attenuation to weak natural VLF radio atmospheric signals.

This attenuation network, combined with an ADC driver IC, provides both the desired signal scaling and the high-current, low-impedance driving that a modern high-resolution delta-sigma ADC requires.

- **Input Current Boosting:** High-resolution 24-bit audio ADCs utilize switched-capacitor input that infrequently draw rapid

spikes of current. The ADC driver acts as a fast-charging buffer to keep the input signal stabilized.

- **Nyquist & Anti-Aliasing Filtering:** Acting as the final hardware boundary before digitization, this stage can be used as a differential amplifier to implement a first-order anti-aliasing filter at 48kHz.
- **Voltage Restrictions & Protection:** The driver must align the common-mode voltage provided by the stereo audio ADC. To completely safeguard the system, low-leakage Schottky or transient-voltage-suppression clipping diodes clamp the output rails, ensuring no stray impulse or surge can ever be amplified to a point where it damages the sensitive ADC silicon.

• 3) Expected Outputs

The sub-circuit requires a technical write-up and simulation on the following:

- **Component Values and Topology:** The U-bridge attenuation circuit requires a set of both series resistors and a set of shunt resistor values (or a potentiometer). Demonstrate that it acts as a controllable attenuator from 0 to -20dBV.
- **Component Selection:** Identify either a FDA integrated circuit or a dedicated ADC driver IC, verifying rail to rail operation. Justify the choice, analyzing Total Harmonic Distortion (THD), Voltage Noise Density, Gain Bandwidth Product, and ensure it works on 3.3V power supplies.
- **SPICE Simulation:** A three stage simulation is required:
  - An AC analysis should be used to ensure the flat passband response, and the desired attenuation values should be verified.
  - A DC transient analysis should be undertaken to prove a voltage spike cannot reach the ADC and damage the input stage.

    - A noise analysis should be made in order to ensure a low noise spectral density, and that the design has a very low noise contribution.

Optional Extra (Stretch Goals):

- Designing a LC lowpass is less necessary with the existing filtering going on, but if all else is completed, try simulating a LC lowpass filter to cut aliasing frequencies while not adding a resistive element to cut input amplitude.